GENEVA GRADUATE INSTITUTE

INSTITUT DE HAUTES ÉTUDES INTERNATIONALES ET DU DÉVELOPPEMENT

GRADUATE INSTITUTE OF INTERNATIONAL AND DEVELOPMENT STUDIES

Typologies, Puzzles, and Parsimony

Practicing Research

James Hollway

Notes

- Causal loop diagram feedback – come see me if you want to talk more!
- Today is a continuation on the theme of *questions* and *concepts*
- My spin on this:
  - again *around* the readings
  - about (I hope useful) *tools*
  - several *examples* from my work

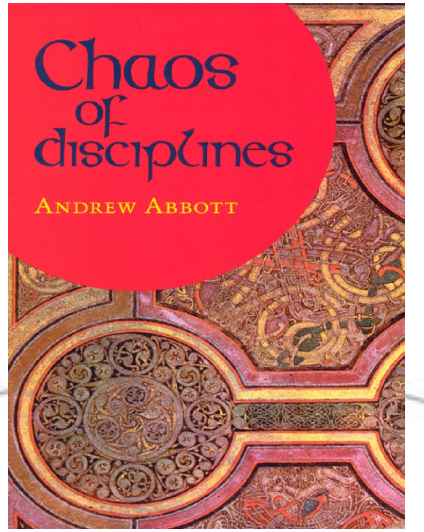
Questions, Puzzles, and Problems

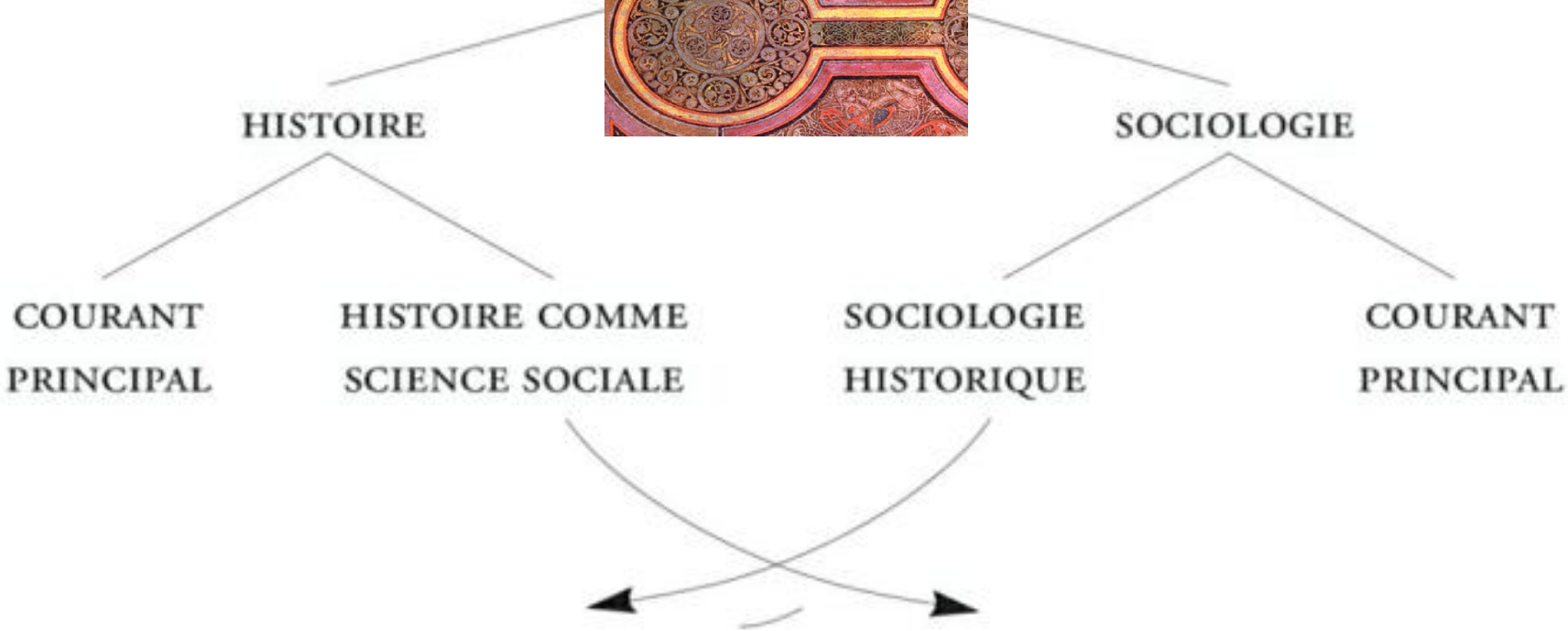
The Gap Trap

- Gaps are…
  - *Unnecessary*: ongoing, unresolved debates can remain fertile ground
  - *Insufficient*: not done before doesn't ensure value (and might signal less value or impossible)
  - *Conservative*: can offer little potential for destablization or innovation
  - *Misleading*: implies complete/sufficient knowledge possible
- Problem-driven research ('great questions of the day') could…
  - Be *fads*: concepts come and go, and hindsight can be embarrassing
  - Make you a *useful idiot*: being driven by others' agendas
  - Make you a *follower*: less fame and influence available in the trails

Sandberg and Alvesson 2011; Gustafsson and Hagström 2018

Chaos of disciplines





Abbott 2001; see also Abbott 2004

Formulating research questions

Avoid … questions	Example	Issue	Instead
no	"International institutions are ineffective"	What do we/you learn if we/you already know the answer?	Ensure a question mark. Italicize to highlight RQ.
multiple	"Are international institutions effective? If so, why? How? How much? … Are they also just? For whom?"	Many (semi-)related questions means many, unconnected answers.	Pick a main question. Leave others as stepping stones/side projects.
binary	"Are international institutions effective?"	Could be answered with yes/no. Too simple. Perhaps a hypothesis?	Use reporter questions, esp. why, how, when, what, etc.
speculative	"How can international institutions be more effective?"	Speculates about possible future. Good MQ but often unanswerable.	Orient RQ towards available evidence that helps inform MQ.
vague	"Why are international institutions bad?"	At least one key concept in question is ambiguous or subjective.	Reconsider meaning. Consider what to look for.
leading	"Why have international institutions been so ineffective?"	Cultivate bias or include unexamined premise.	Rephrase more openly. Examine assumptions.

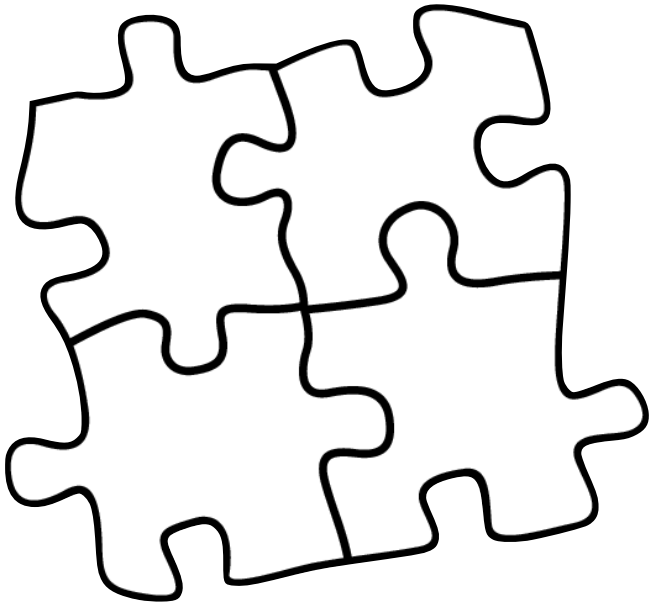
QCVC

How do new states relate to global governance?

Question words	Investigatory type	Answer syntax
What…?	Descriptive	It is… (type)
How…?	Processual	Through… (mechanism)
How much…?	Quantitative	About… (estimate)
When…?	Conjunctive	Under… (conditions)
Why…?	Causal	Because… (explanation/theory)
Why…despite…?	Puzzle	Because… (faulty premise/intervening variable)
..	..	..

This is not exhaustive…

Puzzles

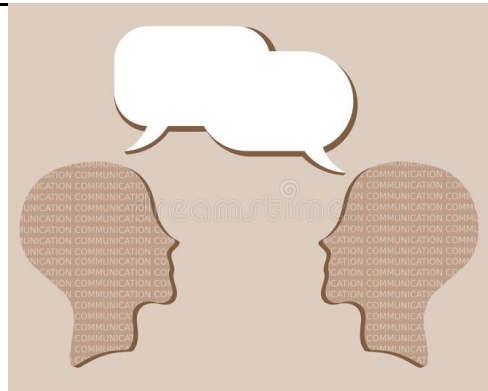


- Why *x* despite *y*? or, How did *x* become possible despite *y*? But see also If *y*, then why *x*?
  - e.g.: why some states go to war, despite complex interdependence, regional integration, or democracy?
- GH's puzzling gap on puzzles:
  - Why so little support for formulating puzzles, despite prescription to identify and resolve puzzles?
  - Really impartial to theoretical approach? (GH 639)
- Require:
  - *Genuinely* puzzling: If claim a puzzle, then it really needs to be a puzzle: check if straightforward explanation among obvious perspectives/theories
    - Avoid common straw puzzles: e.g. is it in the actors self-interest? Is it socially constructed that way? Is it just what the most powerful actor wants?
    - Consider reframing: different explanation as most straightforward, such that puzzle is puzzling and 'new' explanation appears more fruitful a resolution
  - For a particular *audience:* Orients work vis-à-vis state of the art
    - Requires an inventory (or typology??) or previous research
    - Concentrate on the big names first, not because they have it all right, but so that you can relate puzzle to more shared background knowledge

Gustafsson and Hagström 2018

Towards a recipe for constructing research puzzles

1. Make distinctions to narrow down your interest from a topic to an approximation of a research puzzle
2. Explicate the motives and preconceptions underlying your interest in an issue
3. Approach your topic as a political science puzzle, rather than just a political problem
4. Since knowledge is necessary for constructing a research puzzle, read broadly in fields related to your problem area
5. Make sure your research aims to produce new knowledge
6. If the empirical record can be interpreted as conforming to existing assumptions, it is not clear why further research is necessary
7. A gap in previous research is a necessary, but insufficient, argument for new research
8. Problematize the often commonsensical assumptions on which previous knowledge is based
9. The bigger the target, the greater the potential contribution, so determine whether your observation is a case of a wider phenomenon
10. Clarify to what theory, explanation, or interpretation your own thesis would be counterintuitive
11. Construct a clear research puzzle using the puzzle formula 'why *x* despite *y*'
12. Pinpoint a research puzzle early in the research process, but be prepared to find other potentially more significant problems – possibly at higher levels of abstraction – as your knowledge expands



In-class task: Emulate a puzzle-based conversation

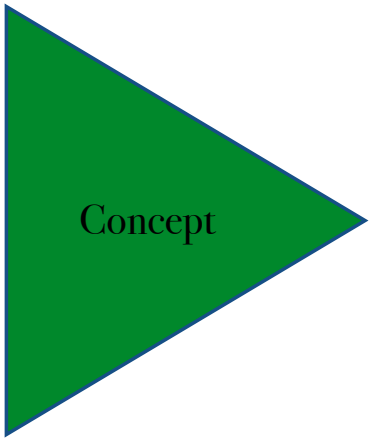
Gustafsson and Hagström 2018

Concepts, Metaphors, And Typologies

QCVC

How do new states relate to global governance?

Term, word,
label, expression



Definition, explanation, meaning,
usage, description, connotation,
intension, property, attribute

Referent, extension, denotation,
phenomena, real-world entity, thing

Ogden-Richards Triangle

Strategies for defining concepts

- **Appeal to authority**: why and to whose authority?

Strategies for defining concepts

- **Appeal to authority**: why and to whose authority?
- **Criterial approach**: under which criteria should we decide between competing definitions or even concepts?

	Gerring (2001)	Gerring (2006)
1	Coherence	Coherence
2	Operationalisation	Differentiation
3	Validity	Depth
4	Contextual Range	Familiarity
5	Resonance	Resonance
6	Field Utility	Field Utility
7	Analytical/empirical utility	Theoretical Utility
8	Parsimony	Parsimony

Strategies for defining concepts

- **Appeal to authority**: why and to whose authority?
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- **Meaning-in-use**: how is it related to other concepts?

Strategies for defining concepts

- **Appeal to authority**: why and to whose authority?
- **Criterial approach**: under which criteria should we decide between competing definitions or even concepts?
- **Meaning-in-use**: how is it related to other concepts?
- **Critical**: what does its use do and whom does it serve?

Try them out on ‘global governance’

QCVC

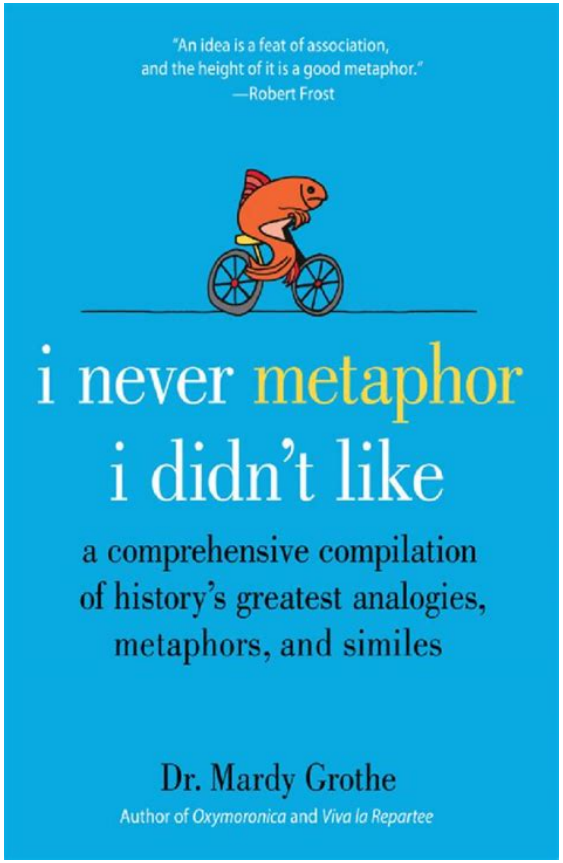
How do new states relate to global governance?

Ladder of interconceptual reasoning	Relationship	E.g.
Simile	Similarity	A like B
Metaphor	1:1	A is B
Analogy	Extended	A is B, which means C
Typology	Conceptual/explanatory	A, B, and C are of X
Framework	Processual/descriptive	X relates to Y
Theory	Causal/constitutive	X causes Y

Ogden-Richards Triangle

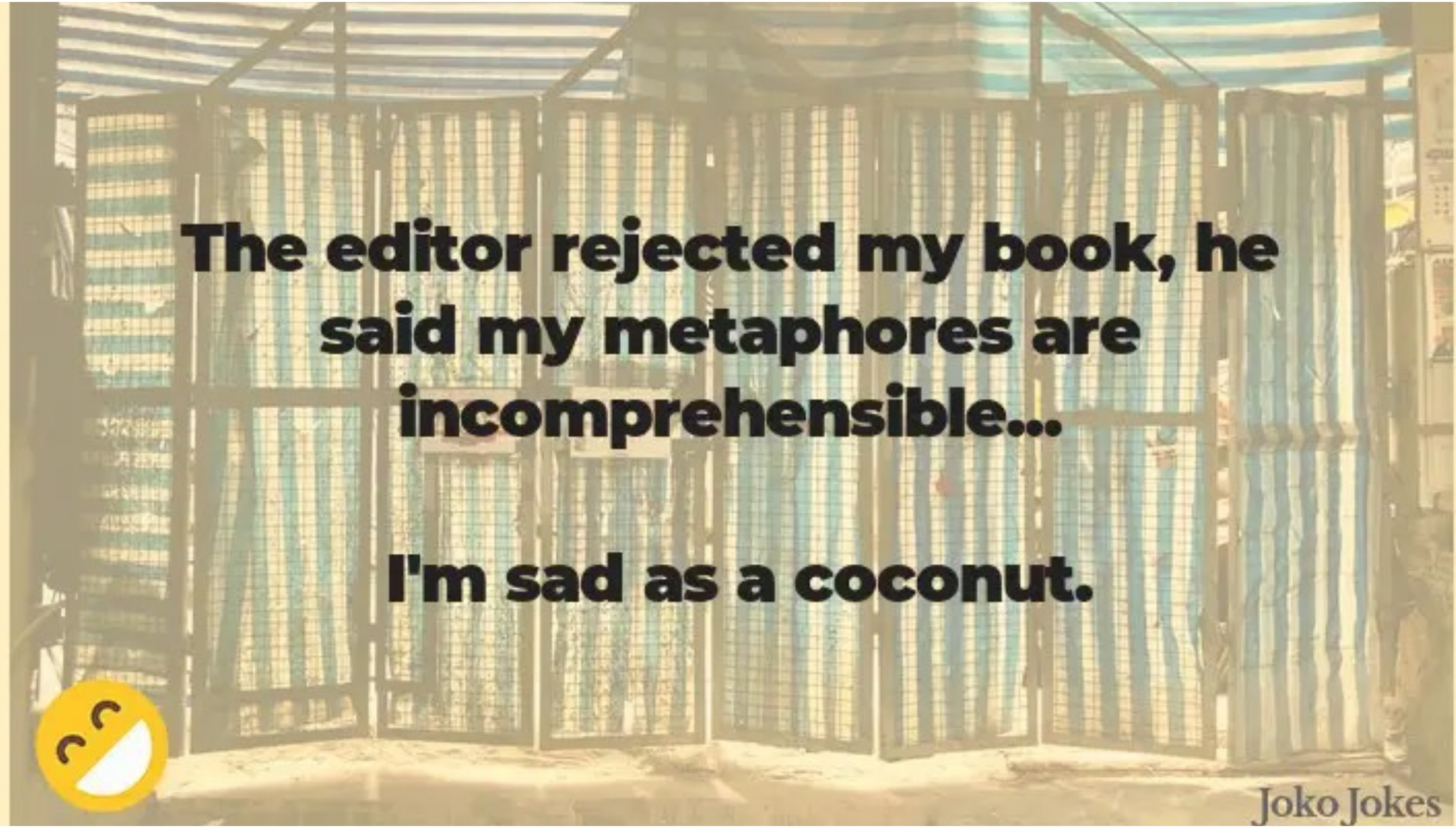
Similes and Metaphors

- What's a simile? It's like a metaphor…
- Transforms the unfamiliar into something familiar/understood, reduces complexity to manageable proportions
- Often quite visual



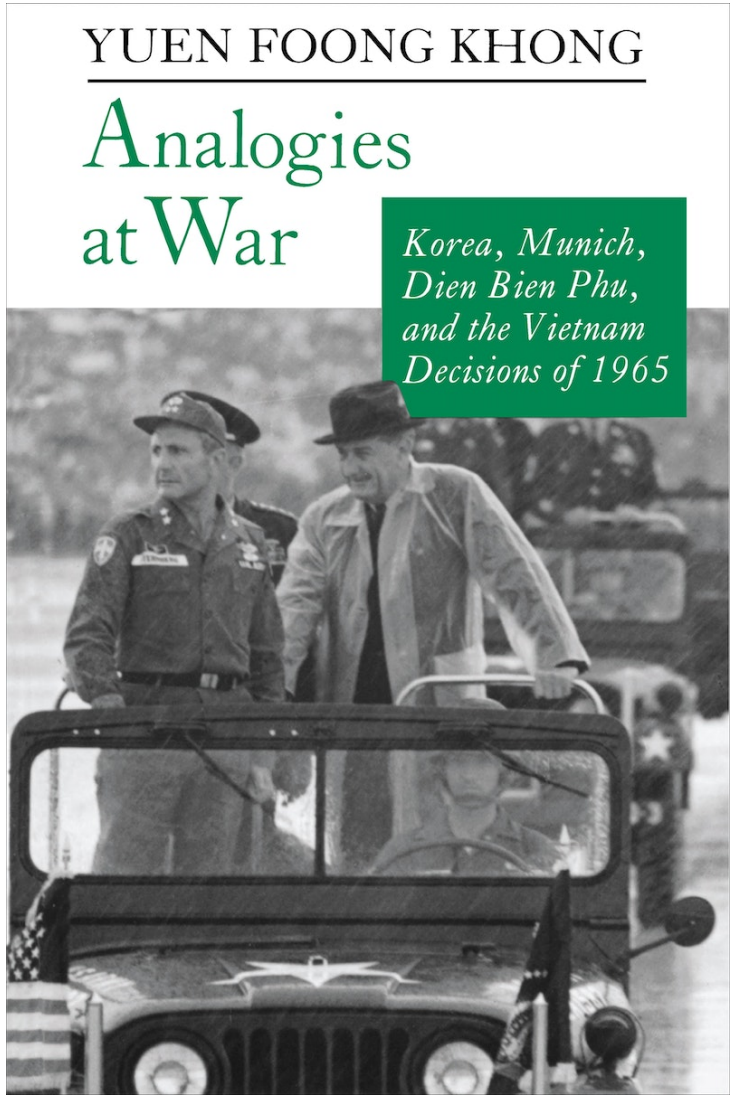
- What does it highlight? What does it hide? What is left out?

Dupont, Hofmann, and Hollway

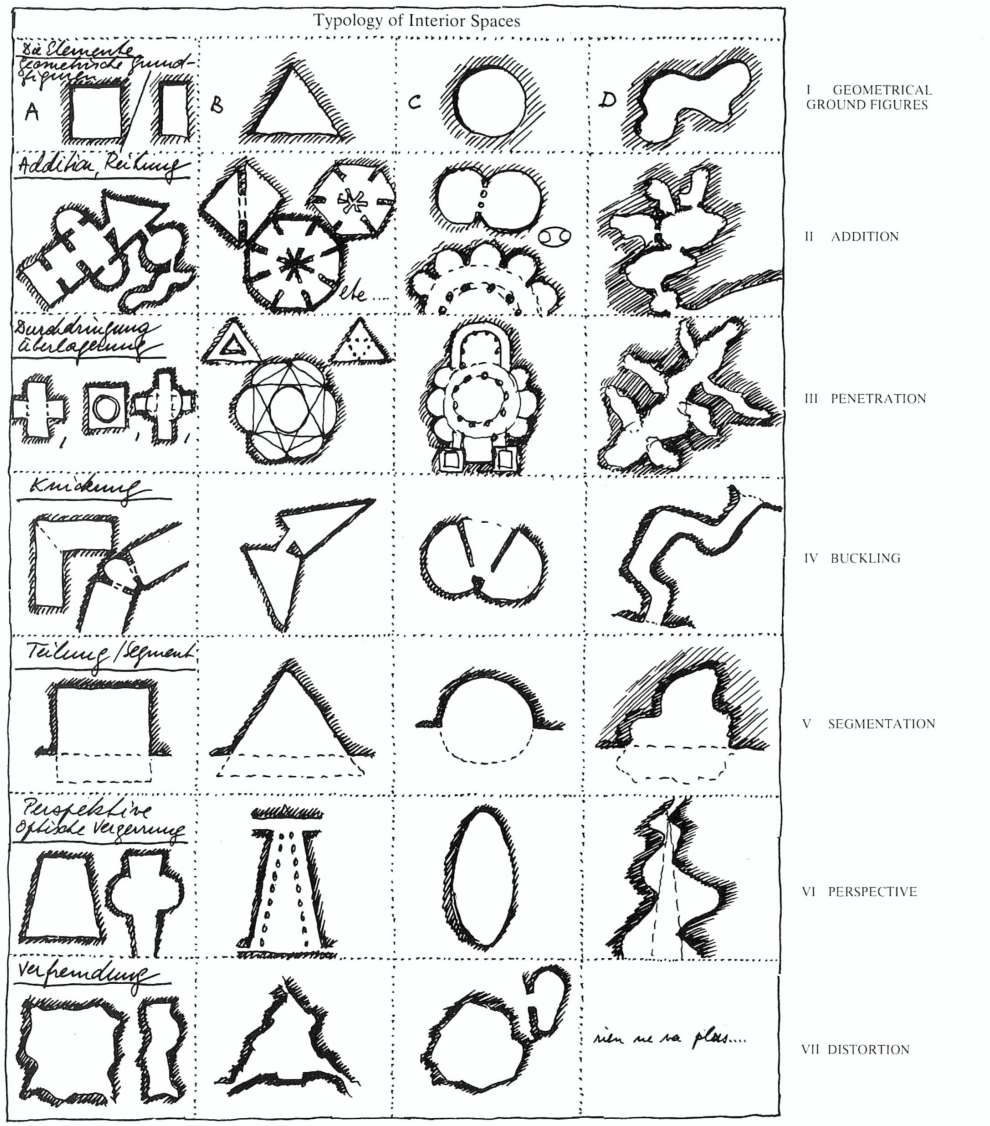

The editor rejected my book, he said my metaphores are incomprehensible...
I'm sad as a coconut.
Joko Jokes

Analogies

- e.g. using Malayan Emergency to reason about Vietnam, and the Vietnam War 'quagmire' to reason about Iraq and others… or Munich to reason about today…
- Decision-makers (as any of us) use analogies as:
  - *Rhetoric*: to justify law and public policy
  - *Reason*: to perform specific cognitive and information-processing tasks essential to decision-making
  - Crucially, they guide action…
- Issues in application:
  - Inaccurate or partially accurate analogies
  - Alternative analogies



Khong 1992



Typologies

- "Finding different manifestations of types of a given phenomenon is the beginning of orderly control and prediction" (Lowi 1972: 299)
- "Language both for formulating the phenomena we seek to explain/ understand and the frameworks we build to explain/understand them" (Berenskoetter 2016: 152)
- One way of pursuing ontological and constitutive theorising (Guzzini 2013)
- Purposes:
  - To address complex phenomena without oversimplifying
  - Clarify similarities and differences among cases to facilitate comparisons
  - Provide a comprehensive inventory of all possible kinds of cases
  - Incorporate interaction effects
  - Draw attention to kinds of cases that have not occurred

Collier et al. 2009; Collier et al. 2012; Bennett and George 2005

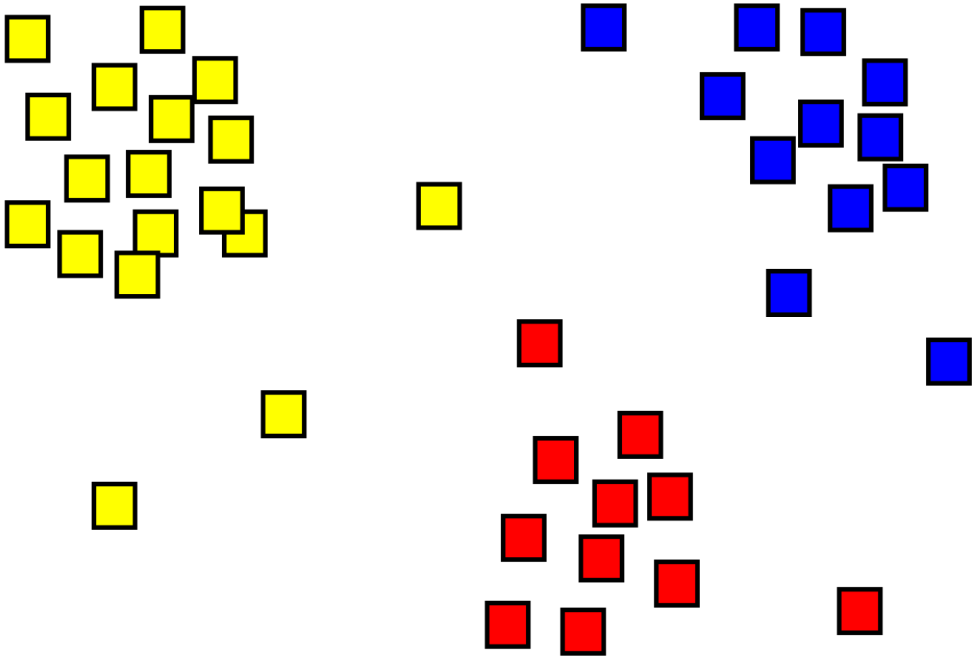
Types of typologies

	Sets	
Dichotomies	$\{0, 1\}$	Single binary category
Scales	$[0, 1]$	Single continuous category
Matrices	$\{\{0, 1\}, \{0, 1\}\}$	Double binary categories
Spaces	$\{[0, 1], [0, 1]\}$	Double continuous categories
Tables	$x, y, z \in A$	Exclusive categories
(Decision) Trees	$x, y \in A; i, j \in B$	Multiple exclusive categories
Venn diagrams	$A \cup B$	Overlapping categories
Ladders and other metaphors	...	...

Incomplete...

How many types?

- Granularity question, depends on:
  - Data:
    - Degrees of freedom
    - Salience
    - Uncorrelated
  - Purpose:
    - To *reason*, *explore*, *explain*, and *communicate* requires simplification
    - To *predict*, *design*, and *act* benefits from high fidelity
      - But be careful of overfitting…
- Inductive pattern recognition
  - Quantitative: e.g. cluster analysis
  - Qualitative: e.g. content analysis or interrater coding



Four galore

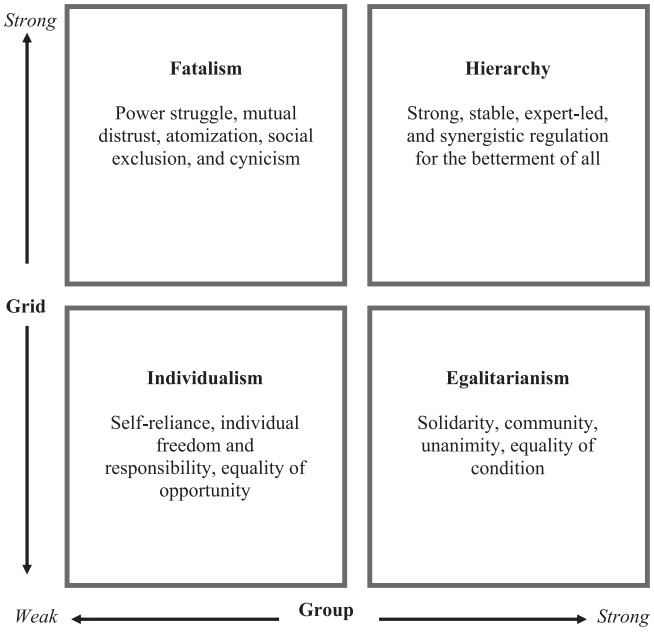


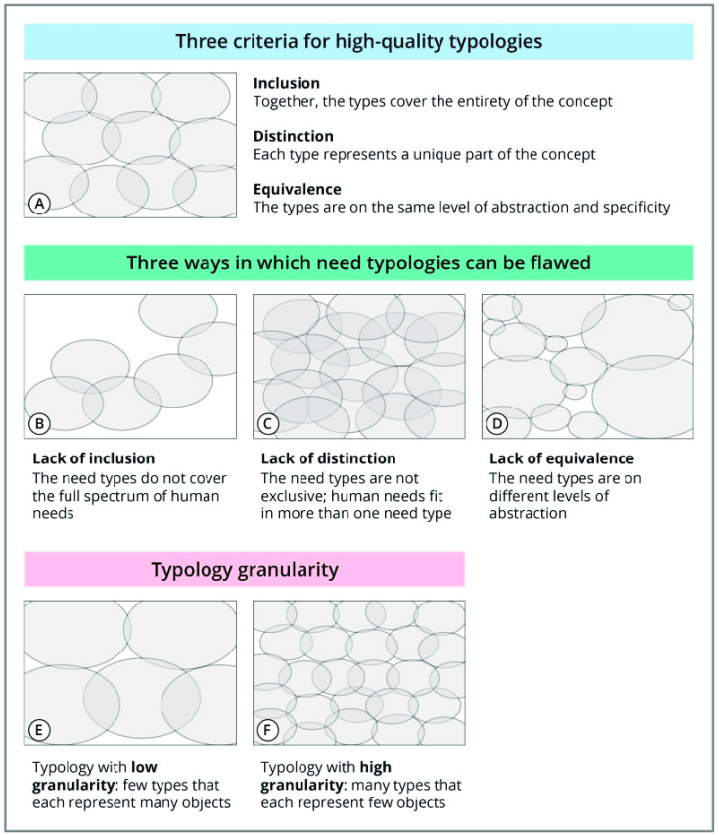
Table 5. Overlap among Cells.

Typology	Group	Grid	Fatalism	Egalitarianism	Hierarchy	Individualism	Sui Generis	No Majority
⋮								
Hirschman (1970)			Neglect	Voice	Loyalty	Exit		
Etzioni (1975) *Compliance*			Coercive	Normative		Utilitarian		Dual
Sartori (1976)	Competitive vs. noncompetitive	Repressive vs. expressive	Unipolar		Monopartism	Polypartism		
Evans (1995)			Predatory		Developmental			
Schwartz (1992)	Self-enhancement/ self-transcendence	Openness to change/ conservation						
Hallin and Mancini (2004)			Polarized pluralist		Democratic corporatist	Liberal		
Wendt (1999)			Hobbesian	Kantian		Lockean		
Triandis (1995); Triandis and Gelfand (1998)			Vertical individualism	Horizontal collectivism	Vertical collectivism	Horizontal individualism		
Williamson (1991)					Hierarchy	Market		Hybrid
Ostrom (2000)				Conditional cooperator	Willing punisher	Rational egoist		
Pierre and Peters (2000)			Network	Community	Hierarchy	Market		
Lindblom (1977)				Persuasion	Authority	Exchange		
Fehr and Schmidt (1999)				Inequity aversion		Selfish		
Schmitter (1974)			Monism	Syndicalism	Corporatism	Pluralism		
Etzioni (1975) *Power*			Coercive	Social normative	Pure normative	Remunerative		
Esping-Andersen (1989)				Social democratic	Corporatist-statist	Liberal		
Etzioni (1975) *Involvement*	Alienative			Social	Pure	Calculative		
Granovetter (1985)			Malfeasance	Trust	Hierarchy	Market		
Ouchi (1980)				Clan	Bureaucracy	Market		
Powell (1990)				Network	Hierarchy	Market		

Weighted agreement <40%.
Weighted agreement ≥40% to <75%.
Weighted agreement ≥75%.

Verweij et al 2020

Topological typologies

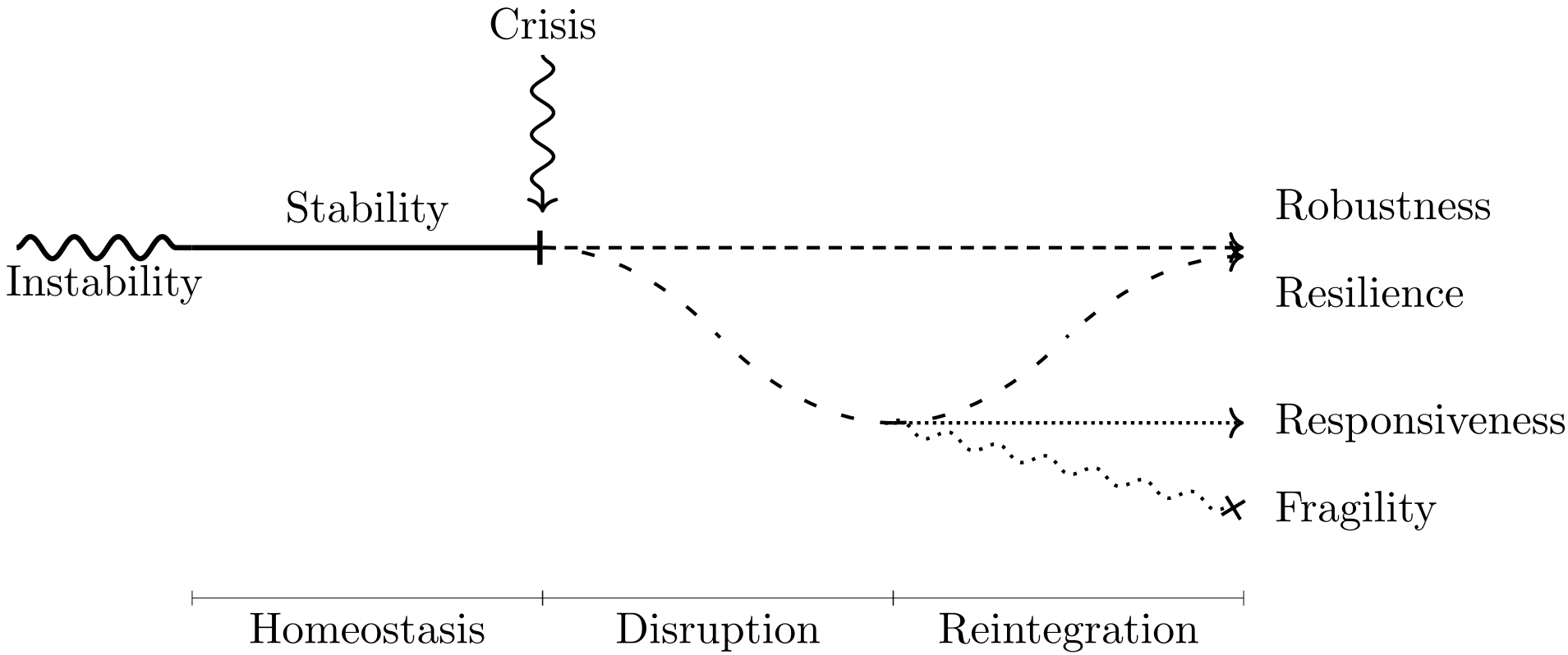


Typology	Inclusion	Distinction
Hooghe and Marks 2003	Low *Wheel not captured, bowl partially captured; type I not relevant*	Low *Hard to differentiate silos from bowls*
Biermann et al 2009	Moderate *Silos not captured*	Low *Hidden integration dimension*
Keohane and Victor 2011	Low *Wheel not captured and integration situation not relevant*	Low *No differentiation within non-hierarchical and loosely coupled situations (bowls and silos)*
Aligica and Tarko 2012 Stephan et al 2019	Moderate-High *Monocentric systems not relevant*	Low *No clear differentiation among polycentric situations*
Eilstrup-Sangiovanni and Westerwinter 2022	Moderate *Scale-variant*	Low-moderate *Two-level structure with ordering mechanisms blurs differentiation*
Henning and Pratt 2022	Moderate-high *Authority restricted to institutional deference*	Moderate *Implicit third dimension (functional/ geographic) blurs differentiation*

Dupont, Hofmann, and Hollway

Framework for resilience of and in networks

	Short	Long
Exogenous	Shock	Shift
Endogenous	Crack	Drift



Hollway 2022; see also Richardson 2002

Ladder of interconceptual reasoning

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Parsimony

- "social science is a forest of falling leaves" (Gunitsky 2019: 711)
- How complex should an explanation be? (Norman 2021: 950)
  - *Occam's razor* vs *law of requisite complexity*
- "F*@! Nuance" vs "Less theory, more description"
- Biomedical research cycle… description → correlation → natural experiments → RCTs

Table 1. Three views of parsimony in social science

Type of parsimony	*Definition*	*Sub type*	*Justification*
Aesthetic	Parsimony as elegance and clarity	Intrinsic	Beauty as an inherent virtue in a theory
		Pragmatic	Clarity and simplicity as aides to theory testing and replication
Ontological	Parsimony as a description of the world	Metaphysical	"God" or "nature" as inherently simplicity-seeking
		Scientific	Symmetry in physical laws means simpler theories are more likely to reflect reality
Epistemological	Parsimony as a stylized theoretical assumption	Cartographic	'Theory as a map'; simplifying assumptions are necessary for useful generalizations
		Empiricist	Parsimony increases falsifiability, improves model design, cuts off the 'garden of forking paths'

Ashby 1958; Hirschman 1984; Lieberman 2016; Healy 2017; Besbris and Khan 2017

This week's assignment

- Draft scripts for an elevator pitch of your project
- Three different audiences: two general IRPS scholars (one positivist leaning, one interpretivist leaning), and a lay family member
- For completion: just a general IRPS scholar
- =<1 page each
- Concentrate on effective set-up (question, problem, puzzle?), metaphorical/analogical reasoning, and relationship between concepts

Some big thoughts for next time

- Lots of questions
- Question the questions
- Is it typologies all the way up?
- How do I relate this to my research design?
- If scholars create metaphors/analogies/typologies through/for own lenses, why should we care?
- When will I be satisfied with the parsimony/nuance of my project?
- In which phase is the literature in my field and how will my work contribute to that?